Yucheng Zhang

Email: yz90@illinois.edu   Mob: +217 819 1275

EDUCATION

University of Illinois Urbana-Champaign **Urbana, IL, USA**
Master of Science in Computer Science (expected July 2027) *Sep. 2025-Jun. 2027*

University of Illinois Urbana-Champaign **Urbana, IL, USA**
Bachelor of Science in Computer Engineering *Sep. 2021-Jun. 2025*

- Cumulative GPA: 3.97/4.0
- Core Courses: Introduction to Algorithms & Models of Computers, Computer Systems Engineering, Probability with Engineering Applications, Machine Learning, IoT and Cognitive Computing

Zhejiang University - University of Illinois Urbana-Champaign Institute **Zhejiang, China**
Bachelor of Engineering in Electrical and Computer Engineering (expected July 2025) *Sep. 2021-Jun. 2025*

- Cumulative GPA: 3.78/4.0
- Core Courses: Data Structures, Computer Systems & Programming, Numerical Analysis, Applied Parallel Programming, Data Science Analytics using Probabilistic Graph Models

Awards and Honors

- **Gold Medal, 46th ICPC Asia Regional Contest Jinan Site** (National) (top 10%) 2021
- Silver Medal, 2021 China Collegiate Programming Contest, Weihai Site (National) 2021
- Dean's List (ZJUI) 2021-2022
- Dean's List (UIUC) 2023-2024

TEACHING EXPERIENCE

CS 101 Intro Computing: Engrg & Sci Teaching Assistant at UIUC Fall 2025 & Spring 2026

- Lead three weekly lab sections
- Host weekly office hours and review sections.
- QA quizs, labs and homework.

MATH 285 Differential Equations Teaching Assistant at ZJUI Spring 2025

- Ran weekly discussion sections, office hours and review sections.
- Graded homework and exams.

MATH 241 Calculus III Teaching Assistant at ZJUI Fall 2024

- Ran weekly discussion sections, office hours and review sections.
- Graded homework and exams.

PUBLICATIONS

- Zimu Zhang*, **Yucheng Zhang*** et.al, "HandX: Scaling Bimanual Motion Generation", CVPR 2026
- Sirui Xu*, Dongting Li*, **Yucheng Zhang*** et.al, "InterAct: Advancing Large-Scale Versatile 3D Human-Object Interaction Generation", CVPR 2025
- Enxin Song*, Wenhao Chai*, Guanhong Wang*, **Yucheng Zhang** et.al, "MovieChat: From Dense Token to Sparse Memory for Long Video Understanding", CVPR 2024

INTERN EXPERIENCE

Internship in Apple   **Advanced Manufacturing Engineer**   Summer 2025

- Start a project about LLM.
- Details cannot be revealed due to Apple Secret Policy.

RESEARCH EXPERIENCE

Bimanual Motion and Interaction Generation | Research Assistant   Apr.2025-Present

Advisor: Yu-Xiong Wang (Assistant Professor, University of Illinois Urbana-Champaign)

- Proposed a unified foundation spanning data, annotation and evaluation for bimanual hand motion.
- Organized the datasets in a unified format to prepare them for annotation pipeline.
- Designed a diffusion-based generation model for bimanual hand motion and developed a unified evaluation pipeline supporting both diffusion and autoregressive models for comprehensive benchmarking.
- Developed a data conversion and interface pipeline to integrate the dataset with physics simulators (e.g., IsaacGym, MuJoCo), enabling physically grounded simulation.
- Paper, “HandX: Scaling Bimanual Motion and Interaction Generation” has been admitted by CVPR 2026.

Human-Object Interaction with SMPL Models | Research Assistant   Feb.2024-Nov. 2024

Advisor: Yu-Xiong Wang (Assistant Professor, University of Illinois Urbana-Champaign)

- Proposed a new method for representing human poses with a small number of markers.
- Built a data collection website on Amazon Mechanical Turk to create a dataset with 22.94 hours of human-object interaction data and detailed annotations.
- Processed data, including splitting video with respect to motion changes by measuring Posecode change and implementing GPT-oriented annotation and augmentation.
- Performed tasks like penetration evaluation and motion prediction on the dataset.
- Paper, “InterAct: Advancing Large-Scale Versatile 3D Human-Object Interaction Generation” has been admitted by CVPR 2025.

Long Video Understanding | Team Leader   Jul. 2023-Dec.2023

Advisor: Gaoang Wang (Assistant Professor, ZJU-UIUC Institute)

- Developed a new memory management method to solve long video understanding problems with high computational load and memory cost.
- Reduced memory costs by storing short-term memory and selectively storing long term memory according to the similarity between frames.
- Built a dataset of 1,000 long videos and manually annotated them with summaries and Q&A.
- Paper, “MovieChat: From Dense Token to Sparse Memory for Long Video Understanding,” accepted by CVPR 2024.

PROJECTS

Smart Cane as Senior Design

- Finish the computer vision part of smart cane alone, use YOLO to find objects and human, use regression to estimate distance and angle, and also recognize the color of traffic lights.
- Finish the GPS part of smart cane, use GPS to navigate the user.
- All code writing in python except YOLO is in C.

Simplified Computer System

- Used C and assembly language to build a simplified computer system that supports various operations, including running programs in parallel, instruction input and storage, and program call.
- Implemented drivers, RTC, system calls, file system, scheduling, memory map, and signals.

Traffic Light Simulation Control System

- Used YOLO to identify vehicles at intersections from videos and created an algorithm to adaptatively change the lasting time of the traffic light according to the presence of waiting cars and accumulated waiting time.
- Built a server to run the YOLO and light control code on FPGA, modified light length according to the cars, and visualized the results with LED lights.

SKILLS AND OTHERS

Programming: MATLAB, C/C++, Python, Sagemath

Frameworks and Tools: Pandas, PyTorch, smplx, Keras, NumPy, SciPy, tqdm